Market Structures and the Distribution of Content

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Motivation

Entertainment products and their markets are unique:

- Both horizontal (genre) and vertical (quality) differentiation.
- Quality is ex-ante uncertain.

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- Main result: these frictions **jointly** lead to market failure.

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- Both horizontal (genre) and vertical (quality) differentiation.
- Quality is ex-ante uncertain.
- Widespread pricing heuristics.
- **Digitization has radically changed these markets.**

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- Main result: these frictions **jointly** lead to market failure.

Model

Products and Producers

Unit mass of producers.

- Producers have idiosyncratic quality $v \sim F$ (independently realized).
- Each producer chooses to produce in a specific genre $g \in \mathcal{G} = \{1, ..., G\}$.
  - A producer creates a good equal to their quality in their chosen genre.

Assumption

F admits a continuous density f with finite mean. Define $\overline{F}(v) = 1 - F(v)$.

Two regimes: producers observe quality either before or after genre choices.

Information Regimes

The information regime changes the strategy space of producers.

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Ex-ante Realization of Quality.
A population strategy for producers corresponds to a mapping $\Psi : \mathbb{R} \to \Delta\mathcal{G}$, where $\Psi(v)$ is the (possibly mixed) strategy of a producer with quality v.

- Ψ induces the measure of content in genre g: $\mu_g^{\Psi}((v, \infty)) = \int_v^{\infty} \Psi_g(v) f(v) dv$.

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Ex-post Realization of Quality.
A population strategy for producers corresponds to a distribution $\psi \in \Delta\mathcal{G}$.

- ψ induces μ^{ψ} the measure of content in genre g: $\mu_g^{\psi}((v, \infty)) = \psi_g \cdot \overline{F}(v)$.

Consumers

There is a unit mass of consumers.

- Each consumer has a type $\mathbf{t} = (t_1, ..., t_G)$:
  - $t_g \in [0, 1]$ determines *how much* content he wants to consume from genre g.
- Let $T \in \Delta\left([0, 1]^G\right)$ be the distribution of types.
- Let T_g be the cdf of demand in genre g, derived from T.
  - $T_g(t)$ is the mass of consumers consuming at most t units from genre g.

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Assumption

T_g admits a continuous density $\frac{dT_g}{dt}(t)$ away from 0, and is full-support on $[0, \bar{t}_g]$, for some $\bar{t}_g \in [0, 1]$.

I allow for $T_g(0) > 0$: some consumers may not consume from genre g.

Consumer Utility and Prices

Consumers receive utility proportional to the quality of consumed goods.

- Consuming measure μ of content yields $\int v d\mu_g(v)$ utility.
  - Constrained to $\mu_g(\mathbb{R}) \leq t_g$.
- Consumers also pay for each good that they consume.

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Focus on two pricing regimes:

Competitive Pricing: a producer of quality v chooses a price in genre g, $p_g(v)$, to maximize profits.

Fixed Pricing: there is a single market price $p_g(v) \equiv p$ for all goods and genres.

- Special case $p = 0$ "Streaming Pricing": producers maximize demand.

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The net utility for a consumer: $v - p_g(v)$ for each product of quality v consumed.

- Consumer $\mathbf{t}$ consumes the t_g units in genre g with the highest *net* utility.

Regimes and Timing.

There are four possible regimes.

- Ex-post vs ex-ante realization of quality.
- Competitive vs fixed pricing.

The timing is as follows:

- Producers choose genres, qualities are realized (information $\Rightarrow$ order).
- If pricing is competitive: Prices are chosen, observing competition.
- Consumers choose bundles to maximize utility.
- Consumer utility and producer profits are realized.

Main Result

Main Result: Both frictions are needed for market failure

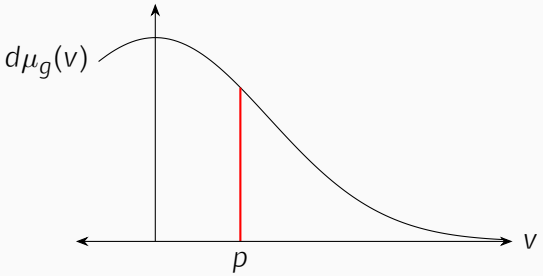
	Ex-ante Realization of Quality	Ex-post Realization of Quality
Competitive Pricing	No Inefficiency Explanation	No Inefficiency Explanation
Fixed Pricing	No Inefficiency* Explanation	**Allocative Inefficiency**

*There is an inefficiency stemming from positive-value products not being consumed: a “Posted Price” inefficiency.

Joint impact of frictions.

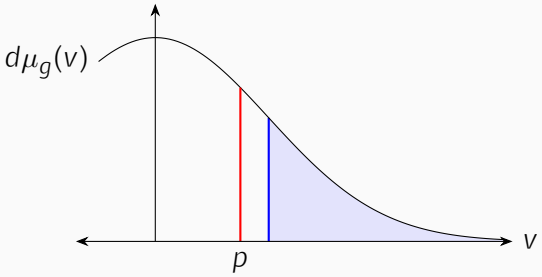
Consumer Utility

Fix the market price p. Since $v - p$ is increasing, consumers will consume the highest quality goods in their genre.



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Consumer $\mathbf{t}$'s genre g utility:

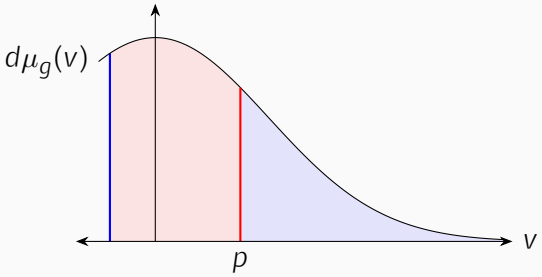
$$\int_{v_g(t_g)}^{\infty} (v - p)\, d\mu_g(v)$$

where $v_g(t_g)$ is such that

$$\mu_g([v_g(t), \infty)) = t_g$$

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$$\mu_g([v_g(t), \infty)) = t_g$$

Producer Profits

Fix an allocation of producers $\psi \in \Delta\mathcal{G}$ and $p \geq 0$.

For a consumer $\mathbf{t}$, facing content μ_g^{ψ}.

- Threshold $v_g(t_g)$ solves $t_g = \psi_g \overline{F}(v)$.

Hence, in genre g, $\mathbf{t}$ consumes everything above $\max\left\{p, \overline{F}^{-1}\left(\frac{t_g}{\psi_g}\right)\right\}$.

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Facing competition ψ, expected profits in genre g are given by

$$\int_0^{\infty} \Pi_p\left(\min\left\{\overline{F}(p), \frac{t}{\psi_g}\right\}\right) dT_g(t), \quad \text{where} \quad \Pi_p(x) \coloneqq \int_{\overline{F}^{-1}(x)}^{\infty} p f(v) dv$$

For $p = 0$, producers maximize expected demand: $\Pi_0(x) \coloneqq \int_{\overline{F}^{-1}(x)}^{\infty} f(v) dv$.

Equilibrium

Definition

An equilibrium ψ^p for market price p is an allocation of producers such that

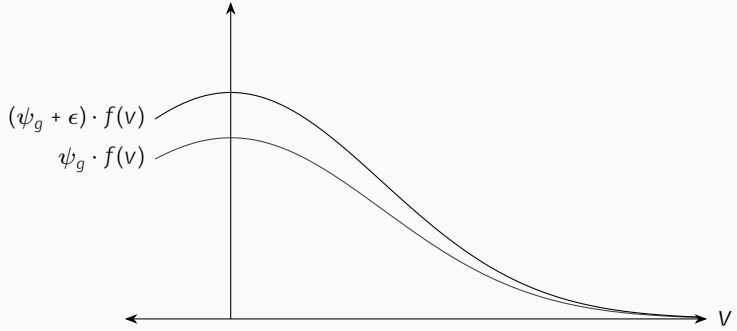
$$\psi_g^p > 0 \Rightarrow g \in \underset{g \in G}{\operatorname{argmax}} \int_0^\infty \Pi_p \left(\min \left\{ \overline{F}(p), \frac{t}{\psi_g^p} \right\} \right) dT_g(t)$$

Lemma

ψ^p is unique for each $p \geq 0$.

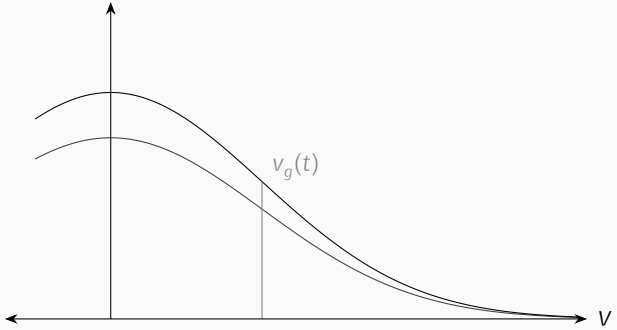
Let $A^p \coloneqq \left\{ g \,|\, \psi_g^p > 0 \right\}$ denote the set of "active" genres with market price p.

Solving the Planner's Problem



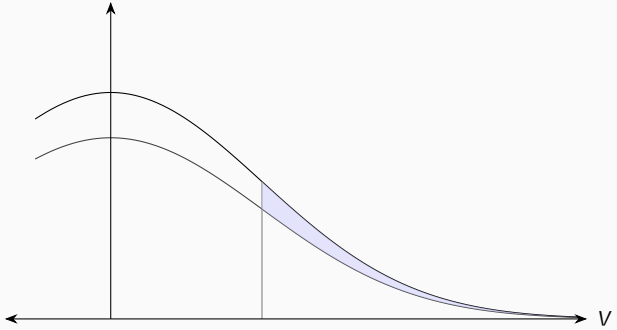
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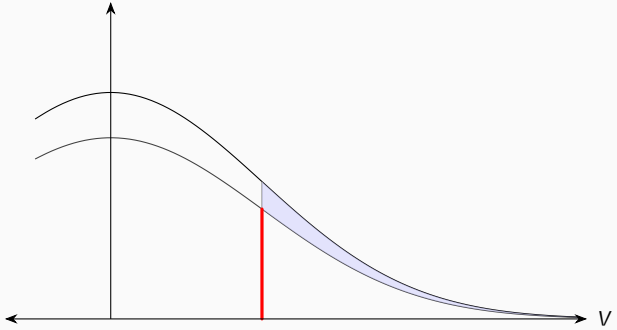
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Consider the benefit from increasing ψ_g:

$$\int_{v_g(t)}^{\infty} \Big(v \qquad \Big) f(v) dv$$

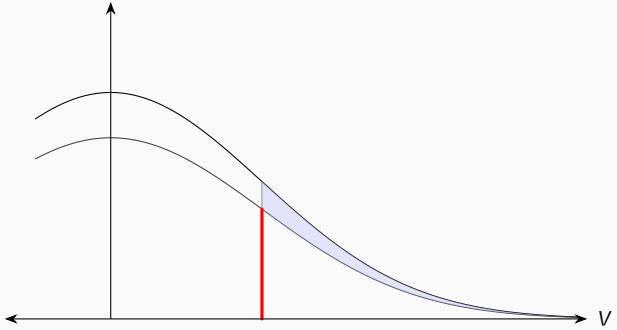
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Consider the benefit from increasing ψ_g:

$$\int_{v_g(t)}^{\infty} \Big(v - v_g(t) \Big) f(v) dv$$

Solving the Planner's Problem



Consider the benefit from increasing ψ_g:

$$M(x) \coloneqq \int_{\overline{F}^{-1}(x)}^{\infty} \left(v - \overline{F}^{-1}(x) \right) f(v) dv$$

Planner's Optimality

The planner hence sets

$$\int_0^\infty M\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g}\right\}\right) dT_g(t)$$

equal across all active g.

Lemma

The socially optimal ψ^* is the unique allocation such that

$$\psi_g^* > 0 \Rightarrow g \in \operatorname*{argmax}_{g \in G} \int_0^\infty M\left(\min\left\{\bar{F}(0), \frac{t}{\psi_g^*}\right\}\right) dT_g(t)$$

Let $A^* := \left\{g \,|\, \psi_g^* > 0\right\}$. This is the set of "active" genres in the efficient allocation.

The Source of the Inefficiency

In the decentralized equilibrium, producers' behaviour sets

$$\int_0^{\infty} \Pi_p \left(\min \left\{ \bar{F}(p), \frac{t}{\psi_g} \right\} \right) dT_g(t)$$

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In the centralized (efficient) allocation, the planner sets

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(Re)statement of Main Result

Theorem

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(ii) If quality is realized ex-post *and* pricing is fixed, there is generically allocative inefficiency above p: $\psi^p \neq \psi^*$.

Genericity is with respect to (F, T), and is required because of edge-cases (e.g., if T_g is equal for all g).

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- Producers' quality rank determines the demand they face. Uncertainty leads to uncertainty over anticipated demand (expectation with respect to dT_g).
- Fixed pricing doesn't affect this ordering (M, Π_p are both increasing), but affects the cardinal trade-off of different demand ($M \neq \Pi_p$).

What does the failure look like?

Range: A measure of concentration

Recall that A^S is the set of active genres in market structure S: $A^S \coloneqq \{g : \psi_g^S > 0\}$.

Definition

Given two market structures, S, S', I say that S *is narrower than* S' if $A^S \subseteq A^{S'}$.

This measure of product range is a coarse measure of concentration: how many genres feature any produced content.

The nature of the failure: Product diversity

Definition

The Hazard rate of $f(v)$ is $H(v) = \frac{f(v)}{\bar{F}(v)}$.

$H(v)$ measures the weight of the tails of the quality distribution.

- Increasing $H(v)$ implies thin tails of f (e.g., Normal, log-concave).
- Decreasing $H(v)$ implies thick tails f (e.g., Pareto).

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(ii) If $H(v)$ is increasing, all competitive equilibria are narrower than the efficient allocation: for all $p \geq 0$, $A^p \subseteq A^*$; and

(iii) If $H(v)$ is decreasing, the efficient allocation is narrower than the streaming equilibrium: $A^* \subseteq A^0$;

Comparative Statics: Changing the value distribution.

Proposition

Fix a transformation $v \mapsto \tau(v)$ with τ increasing and $\tau(0) = 0$. This induces value distribution $\hat{F}(v) = F(\tau^{-1}(v))$.

(i) If τ is concave, then $A^* \subseteq A^*_\tau$;

(ii) If τ is convex, then $A^* \supseteq A^*_\tau$; and

(iii) $A^0 = A^0_\tau$.

- As the tails of F grow, the planner concentrates production in fewer genres.
- This benefit is not internalized by producers.

Comparative statics: Which genres are preferred?

Allow the mass m of producers to vary (baseline model has $m = 1$).

Proposition

Consider two genres g, g' with $T_{g'}$ a mean-preserving spread of T_g.

(i) $\psi_g^p \geq \psi_{g'}^p$, for all p;

(ii) If $H(v)$ is decreasing, $\psi_g^* \geq \psi_{g'}^*$; but

(iii) If $H(v)$ is increasing, there exists an $\overline{m}$ such that if $m > \overline{m}$, $\psi_g^* \leq \psi_{g'}^*$.

- Producers prefer genres with *homogeneous* demand.
- If tails are thin, then a planner prefers to cater to genres with more "super-fans".

Other Results: Quantity-Dependent Pricing and Market Structures

- If producer profits depend convexly on the *quantity* of demand, will concentrate more heavily.
  - E.g., increased royalties after some threshold of sales.

Other Market Structures

- A **monopolist producer** subject to the fixed market price p will be more diverse than the decentralized equilibrium, and possibly more diverse than the efficient allocation.
- An integrated platform using genre-specific **subscription pricing** will be more diverse than optimal.
- A **recommendation system** can improve upon the decentralized equilibrium by committing to suppressing content from genres that are "too popular."

Conclusion

I show how interacting frictions lead to inefficiencies in markets for cultural goods.

- I show that ex-ante quality uncertainty, together with pricing frictions, lead to an inefficient distribution of content.
  - Either friction alone results in efficiency.
- Depending on the distribution of product quality, there can be too much or too little diversity in produced goods.
- In the paper: I explore how changing market structures interact with these frictions.

Epilogue: Generalizing the Framework

This paper highlights how the *interaction* between information an competition leads to market failures. These same frictions exist in many settings with sorting.

- I have another project generalizing these results to a broader context.
- In particular, I study the impact of these frictions in other settings: Workers sorting across sectors, and firms choosing types of R&D projects.
- The takeaway: the interaction of these two frictions lead to sorting inefficiency.

Competitive Pricing, Ex-ante Realization of Quality

- Firms observe their quality, choose their genre, observe competition, and then set prices.

Lemma

There exists a unique pricing equilibrium $p_g(v)$ such that $v - p_g(v)$ is increasing in v.

Proposition

The allocation of producers under competitive pricing and ex-ante realization of quality is efficient.

Back

Competitive Pricing, Ex-post Realization of Quality

The timing is different here: producers choose their genre, observe their quality, and then set their price.

Lemma

For any genre g and mass of producers ψ_g, there exists a unique pricing equilibrium $p_g(v)$ such that $v - p_g(v)$ is increasing in v.

Proposition

The allocation of producers under competitive pricing and ex-post realization of quality is efficient.

Back

Fixed Pricing, Ex-ante Realization of Quality

Consider a fixed price p. If producers know their quality they perfectly anticipate their demand from each genre.

- The highest quality producers will claim the highest demand.

Definition

- Let μ^p be the distribution of quality under the decentralized equilibrium.
- Let μ^* be the socially efficient distribution of producers when quality is realized ex-ante.

They are both unique for producers with positive demand ($v \geq p$).

Proposition

If quality is realized ex-ante, then there is no allocative inefficiency above the market price p: $\mu_g^p(B) = \mu_g^*(B)$ for all $g \in G$ and $B \subseteq (p, \infty)$.